

# Shenni Liang

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## EDUCATION

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- Weill Cornell Medicine, New York City, New York** *May 2029*  
PhD, Tri-Institutional Computational Biology and Medicine (CBM) program
- Cornell University, Ithaca, New York** *May 2024*  
MEng, Computer Science
- Cornell University, Ithaca, New York** *Dec 2023*  
BS, Computer Science

## RESEARCH EXPERIENCE

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- Weill Cornell Graduate School of Medical Sciences, New York City, NY** *Aug 2024 - Present*  
Graduate Student Research Assistant in Prof. Iman Hajirasouliha's Lab
- Build vision models for in vitro fertilization (IVF) embryo time-lapse microscopies
  - Analyze and construct data process pipelines for multi-platform human gut microbial metagenomic samples
- Department of Computational Biology at Cornell University, Ithaca, NY** *Jan 2023 - Jun 2024*  
Undergraduate Student Research Assistant in Prof. Jaehee Kim's Lab
- Developed the agent-based forward-in-time disease outbreak simulation software wrapper in R/Python
  - Established an accessible and user-friendly application for epidemic simulations and studies
  - Designed computational pipelines for executing 5 transmissibility inference methods
  - Investigated effects of confounders (including population structure and drug resistance) on transmissibility inference, showing inferred transmissibility generally exhibits positive correlations with true transmissibility in the absence of any confounders; the accuracy of the methods is influenced by confounders to varying degrees
- Cornell Lab of Ornithology, Ithaca, NY** *Feb 2022 - May 2023*  
Undergraduate Student Research Assistant in Prof. Irby Lovette's Lab
- Analyzed RNA Sequencing data extracted across 5 organs of 18 hummingbirds in R
  - Performed enrichment analysis on differentially expressed genes during hummingbird torpor
  - Identified upregulated pathways (circadian rhythms, blood pressure regulation, and reproductive system development) and downregulated pathways (cell cycle regulation and Keratin metabolism)

## PUBLICATIONS AND MANUSCRIPTS

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Xu, P., **Liang, S.**, Hahn, J., Zhao, V., Lo, W., Haller, B. C., Sobkowiak, B., Chitwood, M., Colijn, C., Cohen, T., Rhee, K., Messer, P., Wells, M., Clark, A. G., Kim, J. (2026) e3SIM: epidemiological-ecological-evolutionary simulation framework for genomic epidemiology. **Methods in Ecology and Evolution** 17(4):1051-1059.

## CONFERENCE PRESENTATIONS

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- Liang, S.**, Hajirasouliha, I. *Vision Model Application on IVF Embryo Development Sequence Images*.  
Talk given at **Computational Genomics Summer Institute (CGSI) 2026**, California
- Xu, P., **Liang, S.**, Haller, B. C., Messer, P. W., Clark, A. G., Kim, J. *Comparing computational methods for inferring Mycobacterium tuberculosis (Mtb) transmissibility from whole genome sequence data*.  
Poster presented at **Ecology and Evolution of Infectious Diseases (EEID) 2023**, Pennsylvania

**Liang, S.**, Blackwell, E. R., Wolfe, S. W., Powers D. R., Shankar A. *Differential gene expression analysis across organs during daily torpor in Anna's Hummingbirds.*

Poster presented at **The Society for Integrative and Comparative Biology (SICB) 2023**, Texas

## AWARDS AND GRANTS

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**Undergraduate Research Experience Fellow**, Cornell Institute of Host-Microbe Interactions and Disease *June 2023*

**Charlotte Mangum Student Support Program**, SICB 2023 *Jan 2023*

**Experiential Learning Grant**, Cornell Lab of Ornithology *May 2022*

**Dean's Honor List for Outstanding Scholastic Performance**, Cornell Engineering *Aug 2020 - May 2023*

## TEACHING EXPERIENCE

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**Operating System Course, Cornell University** *Aug 2023 - Dec 2023*

Undergraduate Student Teaching Assistant

- Hosted weekly office hours for computer processes and concurrent programming queries
- Designed questions for preliminary examinations for approximately 200 students on understanding system calls

**Rising Sophomore Summer Program in Computer Science, Cornell University** *May 2023 - July 2023*

Undergraduate Student Teaching Assistant

- Facilitated Computer System Organization and Programming classes to engage 30 underrepresented students
- Motivated the use of C programming language as a tool for bitmap plotting projects

**Multivariable Calculus Workshop, Cornell University** *Jan 2022 - May 2022*

Undergraduate Student Workshop Facilitator

- Applied active learning strategies to lecture 15 students in an interactive classroom setting
- Created 10 worksheets and weekly session activities to review and preview concepts in engineering math

## WORK EXPERIENCE

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**Nokia Shanghai-Bell, Shanghai, China** *Jun 2021 - Aug 2021*

Front-end Web Development Intern

- Practiced Vue.js to create a budget review website and enabled access control for engineers
- Implemented an interactive web page for displaying live working status of 40 machines in the server room

## LEADERSHIP, SERVICE, AND PROFESSIONAL DEVELOPMENT

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**Tri-I/WCGS Book Club**, Co-president *Aug 2025 - present*

**Professional Development Hub(pd|Hub)**, Participant *Oct 2025 - Dec 2025*

## SKILLS

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**Programming Languages** Python, R, C programming language, java, OCaml, Javascript

**Languages** Mandarin (Expert)